Reliable Classification: Calibration, Uncertainty, and Selective Prediction

April 2026

Abstract

This document is written in three layers. **Part I** is a self-contained first read for a student who knows only basic linear algebra, probability, and calculus. **Part II** gives the deeper mathematical derivations behind the common reliability frameworks. **Part III** gives the historical lineage of the field and a structured survey of influential recent work from 2023–2025.

The central theme is simple: a reliable classifier is not only a classifier that is accurate. It is a system that outputs probabilities, scores those probabilities in a principled way, repairs them when necessary, and then uses them to decide whether to answer or abstain. Accordingly, the note moves from logits and softmax probabilities to calibration metrics, post-hoc calibration, uncertainty toolkits, and finally selective prediction. Historically important works are treated from root to leaves, and recent papers are explained in terms of how they extend the same common framework.

Quick start

If this is your first serious encounter with reliability, read only Part I on the first pass. By the end of Part I you should be able to do eight things: (1) explain calibration in one sentence, (2) distinguish calibration from accuracy and sharpness, (3) compute Brier score, log loss, and ECE on a tiny example, (4) explain why ECE can mislead, (5) fit temperature scaling, (6) explain what deep ensembles and MC dropout do operationally, (7) derive the basic abstention threshold from a reject cost, and (8) read a risk–coverage curve.

How the note is organized

1. **Part I. Core foundations.** Concrete examples first, mathematics second.
2. **Part II. Foundations and full derivations.** Proper scoring rules, Murphy decomposition, advanced post-hoc calibration, deep ensembles, MC dropout, and end-to-end abstention methods.
3. **Part III. History and recent work.** A root-to-leaf map of the field and a survey of influential papers from 2023–2025.

Whenever a section is mainly optional on first read, it is marked explicitly.

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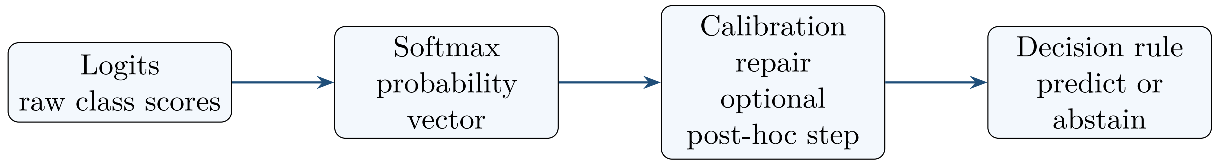

Figure 1: The beginner's mental model for this note. A classifier first produces raw scores, then probabilities, then optionally a calibrated probability vector, and finally an action.

Part I. Core foundations: the self-contained first read

What Part I is trying to teach

A first-time reader often sees too many topics at once and blends them together. Throughout Part I, keep the following six-way distinction in mind.

Category	Plain-English question
Property	Are the probabilities honest? This is *calibration*.
Training loss	What loss rewards honest probabilities? Typical answers: *log loss* and *Brier score*.
Evaluation metric	How do we measure calibration after training? Typical answers: *ECE*, reliability diagrams, NLL, Brier, classwise or local metrics.
Repair method	If the probabilities are not honest, how do we fix them? Typical answers: *temperature scaling*, Platt scaling, isotonic regression, beta calibration, Dirichlet calibration.
Uncertainty toolbox	How do we get a stronger uncertainty signal in the first place? Typical answers: *deep ensembles*, MC dropout.
Decision rule	When should the system answer, and when should it abstain? This is *selective prediction*.

1 Before anything else: the one picture to remember

A modern classifier usually does four things in sequence.

1. It computes *logits*, that is, raw scores for each class.
2. It turns those logits into a probability vector with a softmax.
3. It may apply a *calibration repair* step to make those probabilities numerically more honest.
4. It uses the resulting confidence values to make a decision, which may include abstaining.

That entire chain is reliability. Calibration is only one part of it, but it is the part that gives a numerical meaning to confidence.

Why does this matter? Because in deployment, people do not only look at the top label. They also look at the confidence number attached to that label. If the model says "98%" and the number is not numerically trustworthy, the downstream decision can be wrong even when the top label is often correct on average.

Takeaway after the one-picture view

A classifier used in practice is not only a label generator. It is a *probability estimation system* followed by a *decision system*. Reliability studies that whole chain.

2 The running examples and the data split you must not forget

We will use three tiny examples repeatedly.

2.1 Example A: a binary weather-forecast table

Suppose a forecaster predicts the probability of rain on four days.

Day	Forecast p	Outcome y	Squared error $(p-y)^2$	Log-loss term
1	0.90	1	0.01	$-\log 0.90 \approx 0.105$
2	0.70	0	0.49	$-\log 0.30 \approx 1.204$
3	0.40	1	0.36	$-\log 0.40 \approx 0.916$
4	0.20	0	0.04	$-\log 0.80 \approx 0.223$
Average	–	–	BS = 0.225	NLL $\approx$ 0.612

This table is small enough that we can compute Brier score and log loss by hand. It is therefore the easiest entry point to the idea of *probability quality*.

2.2 Example B: a tiny multiclass neural-network example

Now move to the type of classifier used in deep learning. Suppose the model has three classes, denoted A, B, and C. On each input it outputs a logit vector $z(x) \in \mathbb{R}^3$. The softmax probabilities are

$$p_k(x) = \frac{e^{z_k(x)}}{\sum_{j=1}^{3} e^{z_j(x)}}.$$

Table 1 shows eight toy examples. The probabilities are rounded to three decimals.

#	Logits $z(x)$ and softmax $p(x)$	Pred.	Conf.	True	Correct?
1	$z = (3.5, 1.0, 0.2)$, $p = (0.894, 0.073, 0.033)$	A	0.894	A	1
2	$z = (2.9, 1.7, 0.9)$, $p = (0.696, 0.210, 0.094)$	A	0.696	A	1
3	$z = (1.0, 2.8, 0.7)$, $p = (0.128, 0.777, 0.095)$	B	0.777	B	1
4	$z = (3.0, 2.5, 0.4)$, $p = (0.595, 0.361, 0.044)$	A	0.595	B	0
5	$z = (0.2, 0.8, 2.7)$, $p = (0.067, 0.121, 0.812)$	C	0.812	C	1
6	$z = (3.2, 1.9, 0.5)$, $p = (0.746, 0.203, 0.050)$	A	0.746	B	0
7	$z = (1.8, 1.5, 1.3)$, $p = (0.426, 0.316, 0.258)$	A	0.426	C	0
8	$z = (0.5, 2.2, 2.0)$, $p = (0.091, 0.500, 0.409)$	B	0.500	C	0

Table 1: The multiclass running example used for softmax, ECE, temperature scaling, and selective prediction.

From this table:

$$\text{accuracy} = \frac{4}{8} = 0.5.$$

The average top-label confidence is much larger:

$$\frac{0.894 + 0.696 + 0.777 + 0.595 + 0.812 + 0.746 + 0.426 + 0.500}{8} \approx 0.681.$$

So this toy model is somewhat overconfident: it sounds more certain than its empirical correctness rate suggests.

One explicit softmax calculation. For the first example,

$$z = (3.5, 1.0, 0.2).$$

Then

$$p_A = \frac{e^{3.5}}{e^{3.5} + e^{1.0} + e^{0.2}} \approx \frac{33.115}{33.115 + 2.718 + 1.221} \approx 0.894.$$

The same formula gives $p_B \approx 0.073$ and $p_C \approx 0.033$.

2.3 Example C: same accuracy, different confidence behavior

A beginner's most common misconception is that if two models have the same accuracy, then they are equally reliable. That is false.

Consider the same binary correctness sequence

$$z = (1, 1, 0, 1, 1, 0, 1, 0)$$

but two different confidence sequences:

$$c^{(A)} = (0.95, 0.90, 0.80, 0.75, 0.60, 0.55, 0.40, 0.35),$$

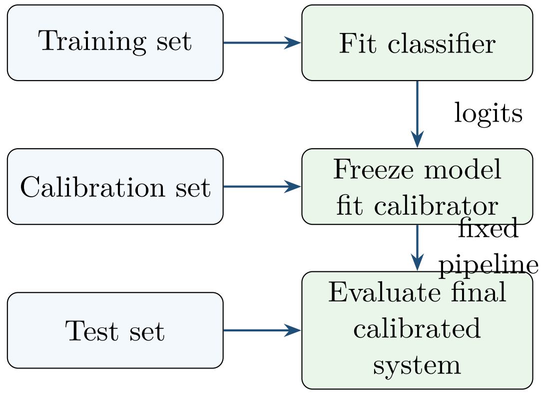

Figure 2: The standard post-hoc calibration pipeline. Train the model first, freeze it, fit the calibrator on a separate calibration set, then evaluate only once on the test set.

$$c^{(B)} = (0.70, 0.70, 0.70, 0.70, 0.55, 0.55, 0.55, 0.55).$$

Both models have the same accuracy, namely 5/8. Yet their probability behavior differs. We will return to this example in the ECE section because it shows that ECE can declare Model B perfect under one binning scheme even though Brier score and log loss still prefer Model A.

2.4 The data split that every novice must memorize

Calibration is not meaningful unless the data split is clean.

The standard split is:

1. **Training set:** fit the original classifier.
2. **Calibration or validation set:** with the classifier frozen, fit the calibrator.
3. **Test set:** evaluate the final calibrated system exactly once.
4. **Deployment data:** future data, which may differ from the validation or test distribution.

Common beginner mistake

Do not fit the calibrator on the test set. That leaks information and makes the reported reliability too optimistic. A surprisingly large amount of confusion in the literature disappears once this split discipline is respected.

Takeaway after the examples

We now have all the concrete objects we need: binary probabilities, multiclass logits and softmax outputs, top-label confidence, correctness indicators, and a clean train/calibration/test split. The rest of Part I simply explains what these objects mean and how they are used.

3 What calibration means

3.1 Calibration in plain words

Calibration asks whether a probability statement is numerically honest.

If a model says "0.8", then among all examples receiving the prediction 0.8, roughly 80% should truly be positive. That sentence is the entire idea.

Formally, for binary classification with a predicted probability $\hat{p}(X) \in [0,1]$,

$$\mathbb{P}(Y = 1 \mid \hat{p}(X) = p) = p.$$

That is a population statement. In real datasets we do not usually observe infinitely many points with exactly the same predicted value, so we approximate this ideal with bins or smoothing.

3.2 Three levels: binary, top-label, and full multiclass

There are three notions worth separating.

Definition 3.1 (Binary calibration)**.** *A binary predictor $\hat{p}(X)$ is calibrated if*

$$\mathbb{P}(Y = 1 \mid \hat{p}(X) = p) = p$$

for every attainable value p.

Definition 3.2 (Top-label or confidence calibration)**.** *For multiclass probabilities $p(x) = (p_1(x), \dots, p_K(x))$, let*

$$\hat{y}(x) = \arg\max_k p_k(x), \qquad c(x) = \max_k p_k(x).$$

The predictor is confidence-calibrated if

$$\mathbb{P}(Y = \hat{y}(X) \mid c(X) = c) = c.$$

Definition 3.3 (Classwise or full-vector calibration)**.** *A stronger notion asks, for each class k, whether*

$$\mathbb{P}(Y = k \mid p_k(X) = p) = p.$$

This uses the entire probability vector, not only the top confidence.

3.3 Calibration versus accuracy

Accuracy asks whether the *largest* probability corresponds to the correct label. Calibration asks whether the *value* of that probability is honest.

The difference is easier to see through Example C. Model A and Model B have the same accuracy. But if a model regularly says 0.95 and is correct only half the time, then the number 0.95 should not be trusted. A model can therefore be accurate and overconfident.

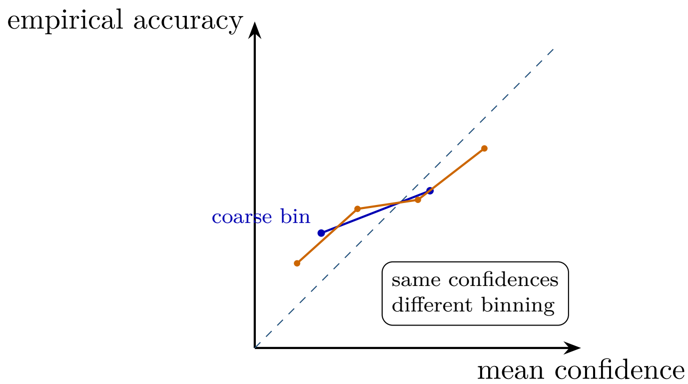

Figure 3: The same toy confidences grouped in two different ways. Because the bins differ, the apparent calibration picture changes. This is why ECE must always be interpreted together with its binning rule.

3.4 Calibration versus sharpness or resolution

Calibration is not the same thing as being informative.

A predictor that always outputs 0.50 in a balanced binary problem can be perfectly calibrated. But it is not useful, because it never distinguishes easy cases from hard cases.

This motivates a second concept.

Definition 3.4 (Sharpness or resolution)**.** *Sharpness asks how much the model separates different inputs by confidence. A sharp predictor uses values such as 0.95, 0.75, 0.20, and 0.05 in a meaningful way instead of collapsing everything near the global base rate.*

Hence:

- **Accuracy** is about the top class.
- **Calibration** is about the honesty of the probabilities.
- **Sharpness** is about whether the model meaningfully separates easy and hard cases.

A full reliability report must consider all three.

3.5 Reliability diagrams

A reliability diagram compares mean confidence in each bin to empirical accuracy in that bin. The diagonal line means perfect calibration.

If a point lies below the diagonal, confidence exceeds accuracy and the model is overconfident in that region. If a point lies above the diagonal, the model is underconfident there.

Takeaway after the definition section

Calibration is about the numerical meaning of probabilities. Accuracy is not enough, and perfect calibration alone is not enough either because a model may still be uninformative. The useful mental picture is: *honesty* plus *informativeness*.

Checkpoint questions

Can you answer the following before moving on?

1. In one sentence, what is calibration?
2. Why can two models have the same accuracy but different calibration?
3. Why can a perfectly calibrated model still be unhelpful?

4 Losses that reward honest probabilities

A novice often asks: "If calibration is about honest probabilities, what loss should I use when I train or evaluate those probabilities?"

The classical answer is: use a *proper scoring rule*. For a first read, the easiest way to understand this idea is through the Brier score and log loss.

4.1 Brier score

For binary labels $y \in \{0, 1\}$ and a predicted probability $a \in [0, 1]$, the Brier score is

$$S_{\mathrm{Br}}(a, y) = (a - y)^2.$$

It is small when the probability is close to the outcome and large when it is far away.

On Example A, the average Brier score is

$$\mathrm{BS} = \frac{0.01 + 0.49 + 0.36 + 0.04}{4} = 0.225.$$

Why the Brier score targets the true conditional probability. Fix an input x and write

$$\eta(x) = P(Y = 1 \mid X = x).$$

Suppose we announce some number a. The conditional expected Brier loss is

$$L_{\mathrm{Br}}(a) = \eta(x)(1 - a)^2 + (1 - \eta(x))a^2.$$

Expand:

$$L_{\mathrm{Br}}(a) = \eta(x) - 2\eta(x)a + \eta(x)a^2 + a^2 - \eta(x)a^2 = \eta(x) + a^2 - 2\eta(x)a.$$

Differentiate:

$$\frac{dL_{\mathrm{Br}}}{da} = 2a - 2\eta(x).$$

Setting the derivative to zero gives

$$a = \eta(x) = P(Y = 1 \mid X = x).$$

The second derivative is

$$\frac{d^2 L_{\mathrm{Br}}}{da^2} = 2 > 0,$$

so this point is the unique minimizer. Thus the Brier score is minimized by the true conditional probability.

4.2 Log loss

For a binary forecast a and outcome y,

$$S_{\log}(a, y) = -y \log a - (1 - y) \log(1 - a).$$

On Example A, the average log loss is approximately

$$\mathrm{NLL} \approx 0.612.$$

Why log loss also targets the true conditional probability. Again write $\eta(x) = P(Y = 1 \mid X = x)$. The conditional expected log loss at x is

$$L_{\log}(a) = -\eta(x) \log a - (1 - \eta(x)) \log(1 - a).$$

Differentiate:

$$\frac{dL_{\log}}{da} = -\frac{\eta(x)}{a} + \frac{1 - \eta(x)}{1 - a}.$$

Set the derivative to zero:

$$-\frac{\eta(x)}{a} + \frac{1 - \eta(x)}{1 - a} = 0 \quad \Longrightarrow \quad \eta(x)(1 - a) = a(1 - \eta(x)).$$

Expanding both sides gives

$$\eta(x) - \eta(x)a = a - a\eta(x).$$

The mixed terms cancel, leaving

$$a = \eta(x).$$

The second derivative is

$$\frac{d^2 L_{\log}}{da^2} = \frac{\eta(x)}{a^2} + \frac{1 - \eta(x)}{(1 - a)^2} > 0,$$

so this is again the unique minimizer.

4.3 What "proper scoring rule" means

The last two derivations illustrate the general definition.

Definition 4.1 (Proper scoring rule). *A scoring rule is a loss assigned to a probability forecast and an outcome. It is* proper *if the true probability distribution minimizes expected score. It is* strictly proper *if the minimizer is unique.*

In plain language: a proper scoring rule rewards *honest* probabilities.

4.4 Why ECE alone is not enough

A model can achieve a small ECE under a particular binning scheme and still have a worse probability distribution under Brier score or log loss. Example C is a simple illustration. Model B can have ECE equal to zero under a coarse binning rule, yet Model A still achieves a better Brier score and a better log loss because it preserves more useful confidence variation.

This is why reliable reporting usually includes:

- a proper score such as log loss or Brier score,
- a descriptive calibration statistic such as ECE,
- and a reliability diagram.

Takeaway after the loss section

Brier score and log loss matter because they are minimized by the true conditional probability. That is the theoretical root of probability estimation. ECE, by contrast, is only a descriptive summary of calibration error. It is useful, but it is not the object that defines truthful probability estimation.

5 Expected Calibration Error (ECE): how to compute it and why it can mislead

5.1 The step-by-step definition

For top-label calibration, the inputs to ECE are:

- the confidence values c_i,
- the correctness indicators $z_i = \mathbf{1}\{\hat{y}_i = y_i\}$,
- and a chosen set of bins.

Suppose the bins are $B_1, \dots, B_M$. For each bin B_m, define:

$$\mathrm{acc}(B_m) = \frac{1}{|B_m|} \sum_{i \in B_m} z_i, \qquad \mathrm{conf}(B_m) = \frac{1}{|B_m|} \sum_{i \in B_m} c_i.$$

Then

$$\mathrm{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{n} \left|\mathrm{acc}(B_m) - \mathrm{conf}(B_m)\right|.$$

This is a weighted average of binwise calibration gaps.

5.2 Computing ECE on the multiclass running example

For Example B, take three equal-width bins:

$$[0, \tfrac{1}{3}), \qquad [\tfrac{1}{3}, \tfrac{2}{3}), \qquad [\tfrac{2}{3}, 1].$$

The confidences are

$$(0.894, 0.696, 0.777, 0.595, 0.812, 0.746, 0.426, 0.500),$$

and the correctness indicators are

$$(1, 1, 1, 0, 1, 0, 0, 0).$$

The first bin contains no points. The second bin contains confidences

$$0.595,\ 0.426,\ 0.500$$

with correctness

$$0,\ 0,\ 0.$$

So

$$\mathrm{acc} = 0, \qquad \mathrm{conf} = \frac{0.595 + 0.426 + 0.500}{3} \approx 0.507.$$

Its contribution to ECE is

$$\frac{3}{8} \cdot |0 - 0.507| \approx 0.190.$$

The third bin contains the remaining five points, with empirical accuracy

$$\frac{4}{5} = 0.8$$

and average confidence

$$\frac{0.894 + 0.696 + 0.777 + 0.812 + 0.746}{5} \approx 0.785.$$

Its contribution is

$$\frac{5}{8} \cdot |0.8 - 0.785| \approx 0.009.$$

Hence

$$\mathrm{ECE} \approx 0.190 + 0.009 = 0.199.$$

5.3 A practical algorithm for ECE

Given confidence–correctness pairs (c_i, z_i):

1. choose the bins;
2. place each example into its bin;
3. compute mean confidence and mean correctness in each nonempty bin;
4. take the absolute gap in each bin;
5. average those gaps with weights proportional to bin size.

That is all ECE does.

5.4 Why ECE became popular

ECE became popular because it is easy to explain, easy to compute, and easy to plot. For a first pass through the literature, that popularity is understandable. But simplicity also creates several pitfalls.

5.5 Pitfall 1: ECE depends on the binning scheme

Figure 3 already showed the visual version of this issue. The numerical version is equally important. For the same Example B:

- with three equal-width bins, ECE ≈ 0.199;
- with four equal-mass bins, ECE ≈ 0.254.

The model did not change. Only the bins changed.

5.6 Pitfall 2: ECE is usually a biased estimator

ECE is computed after grouping points into bins and replacing each point by a bin average. That coarse grouping introduces estimation bias. Roelofs *et al.* [43] study this carefully and propose debiasing ideas.

For a beginner, the main lesson is simpler: *ECE is not a gold-standard population quantity that we observe exactly. It is a noisy and bin-dependent estimate.*

5.7 Pitfall 3: top-label ECE sees only one number per example

If an application only displays the top confidence to a human, then top-label ECE may be appropriate. But if a downstream system uses the full probability vector, then top-label ECE hides potentially important errors in the non-top classes.

5.8 Pitfall 4: ECE can hide classwise or subgroup failures

A global average can look small even when one class or one subgroup is badly calibrated. This is one reason the literature moved toward classwise, local, and subgroup notions of calibration.

5.9 Pitfall 5: ECE is not a proper scoring rule

ECE does not define truthful probability estimation in the way Brier or log loss do. It is a descriptive statistic. That means it is useful for diagnosis but dangerous as the only reported number.

5.10 Pitfall 6: ECE can reward trivial blandness

Return to Example C. Under one coarse two-bin scheme, Model B obtains ECE $= 0$, while Model A obtains a nonzero ECE. Yet Model A has better Brier score and better log loss:

$$\mathrm{BS}_A = 0.2075 < 0.2213 = \mathrm{BS}_B,$$

$$\mathrm{NLL}_A \approx 0.589 < 0.633 \approx \mathrm{NLL}_B.$$

Model B looks better only because it collapses many distinct confidence values into a bland average that happens to match the bin average accuracy. This is precisely the kind of failure that recent papers criticize [3, 18].

5.11 What a good report should include

A reliable beginner report should not say only "ECE is 0.03". It should say at least:

1. which calibration notion is being evaluated: top-label, classwise, or full-vector;
2. which ECE variant is used: equal-width, equal-mass, classwise, debiased, and with how many bins;
3. a proper score such as NLL or Brier;
4. a reliability diagram;
5. the validation/test split on which the metric is computed.

Takeaway after the ECE section

ECE is a useful descriptive summary, not a complete theory of calibration. Always ask: What bins were used? Which calibration notion is being measured? What do NLL or Brier say? Does the reliability diagram tell the same story?

Checkpoint questions

Can you now explain:

1. how to compute ECE from a tiny table;

2. why ECE changes when the bins change;
3. why a small ECE alone does not guarantee good probability estimation?

6 Post-hoc calibration: temperature scaling first

A *post-hoc calibrator* takes a trained classifier as fixed and learns a new map from its outputs to better probabilities. The data split from Figure 2 is therefore essential: the base model is frozen, and only the calibrator is trained on the calibration set.

6.1 The generic post-hoc calibration problem

Write the frozen model's output as either a score vector $s(x)$ or a probability vector $p(x)$. A post-hoc calibrator learns a map

$$g: \text{ old outputs} \longrightarrow \text{new probabilities}$$

by minimizing some calibration objective on a holdout set:

$$\arg\min_{g \in \mathcal{G}} \sum_{i=1}^{m} \ell(g(s_i), y_i).$$

Different methods choose different families $\mathcal{G}$.

6.2 Why temperature scaling became the default baseline

Temperature scaling [17] acts on logits:

$$p_k^{(T)}(x) = \frac{e^{z_k(x)/T}}{\sum_{j=1}^{K} e^{z_j(x)/T}}, \qquad T > 0.$$

It is popular because:

- it has only one parameter;
- it is easy to fit;
- it preserves the top-class ranking;
- and in many deep-learning settings it works surprisingly well.

What the temperature does.

- If $T > 1$, the distribution becomes softer and confidence values move toward the middle.
- If $T < 1$, the distribution becomes sharper and confidence values move toward 0 or 1.
- Because every logit is divided by the same positive number, the class ordering does not change.

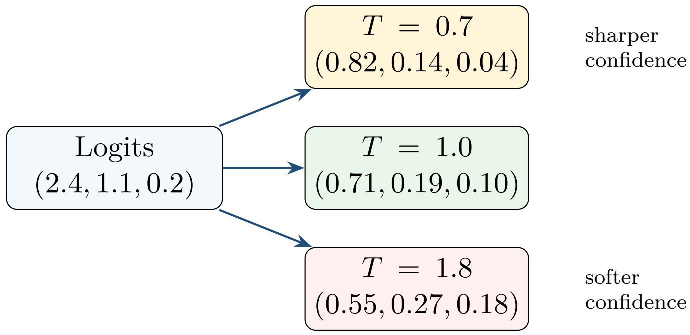

Figure 4: Temperature scaling changes the *shape* of the softmax distribution but preserves the class ranking. Higher temperature softens the probabilities; lower temperature sharpens them.

Figure 4 illustrates the effect on one logit vector.

6.3 Full derivation of the temperature-scaling objective

Suppose the calibration set contains logits and labels

$$\{(z_i, y_i)\}_{i=1}^{m}.$$

Fit T by minimizing calibration-set negative log-likelihood:

$$\mathcal{L}(T) = -\sum_{i=1}^{m} \log p_{i,y_i}^{(T)} = \sum_{i=1}^{m} \left(-\frac{z_{i,y_i}}{T} + \log \sum_{k=1}^{K} e^{z_{i,k}/T} \right).$$

The cleanest way to analyze the objective is to reparameterize by

$$s = \frac{1}{T}.$$

Then

$$\mathcal{L}(s) = \sum_{i=1}^{m} \left(-s z_{i,y_i} + \log \sum_{k=1}^{K} e^{s z_{i,k}} \right).$$

Step 1: compute the gradient. Define

$$p_{i,k}^{(s)} = \frac{e^{s z_{i,k}}}{\sum_{j=1}^{K} e^{s z_{i,j}}}.$$

Differentiate:

$$\frac{d\mathcal{L}}{ds} = \sum_{i=1}^{m} \left(-z_{i,y_i} + \sum_{k=1}^{K} p_{i,k}^{(s)} z_{i,k} \right).$$

Step 2: compute the second derivative. Differentiate again:

$$\frac{d^2\mathcal{L}}{ds^2} = \sum_{i=1}^{m} \left(\sum_{k=1}^{K} p_{i,k}^{(s)} z_{i,k}^2 - \left(\sum_{k=1}^{K} p_{i,k}^{(s)} z_{i,k} \right)^2 \right).$$

The bracketed term is the variance of the logits under the softmax weights $p_i^{(s)}$. So

$$\frac{d^2\mathcal{L}}{ds^2} = \sum_{i=1}^{m} \mathrm{Var}_{p_i^{(s)}}[z_{i,k}] \geq 0.$$

Step 3: conclude convexity. Because the second derivative is nonnegative, the objective is convex in $s = 1/T$. So standard one-dimensional optimization finds the global optimum.

6.4 Why temperature scaling preserves the top class

Suppose class a has the largest original logit:

$$z_a(x) \geq z_b(x) \qquad \text{for all } b.$$

Dividing by a positive number T preserves the inequality:

$$\frac{z_a(x)}{T} \geq \frac{z_b(x)}{T}.$$

Softmax is monotone in each coordinate, so the same class remains largest after scaling. Therefore temperature scaling changes the *numerical value* of confidence, but not the predicted class or the score ranking.

This fact is both a strength and a limitation: temperature scaling can improve numerical honesty, but by itself it does not improve ranking quality for selective prediction.

6.5 The basic algorithm

Algorithm: temperature scaling

1. Train the base classifier on the training set.
2. Freeze the classifier.
3. Collect logits z_i and labels y_i on a separate calibration set.
4. Minimize calibration-set NLL over one scalar parameter $T > 0$.
5. On test or deployment data, replace $\mathrm{softmax}(z)$ by $\mathrm{softmax}(z/T)$.

6.6 The rest of the post-hoc family in one page

Temperature scaling is the most common baseline, but it is not the whole family.

Method	Main idea	When it is useful
Platt scaling [38]	Learn a logistic map $s \mapsto \mathrm{sigmoid}(As + B)$.	Binary or one-vs-rest settings; historically important baseline.
Isotonic regression [46]	Learn a monotone but otherwise flexible piecewise-constant map.	When enough calibration data exist and you want flexibility rather than a one-parameter map.
Beta calibration [21]	Learn a logistic map on $\log s$ and $\log(1 - s)$.	A stronger binary post-hoc family than plain Platt scaling.
Dirichlet calibration [22]	Recalibrate the full multiclass probability vector using log-probability features.	When the whole vector matters, not only the top confidence.

The deeper derivations of these methods are in Part II. For a beginner, it is enough to remember that they all solve the same abstract problem: *learn a map from old scores to better probabilities on a clean holdout set.*

What temperature scaling does *not* fix

Because temperature scaling is monotone in the logits, it does not change score ordering. So if the model ranks examples badly for abstention or retrieval, temperature scaling alone will not repair that ranking.

Takeaway after post-hoc calibration

Post-hoc calibration means: freeze the model, fit a separate map on a clean holdout set, and change the probabilities without retraining the classifier. Temperature scaling is the default baseline because it is simple, convex after reparameterization, and often effective.

7 The practical reliability toolbox

A practitioner often wants a short answer to the question: "What should I actually try first?"

The three most common practical tools are temperature scaling, deep ensembles, and MC dropout.

7.1 A short operational summary

Tool	What you actually do	Intuition
Temperature scaling	Learn one scalar T on a calibration set.	Keep the model fixed but make the probabilities numerically more honest.
Deep ensembles	Train several models independently and average their probabilities.	Approximate model averaging by diversity across different trained networks.
MC dropout	Keep dropout active at test time, run several forward passes, and average the probabilities.	Approximate averaging over many subnetworks sampled by dropout masks.

7.2 Deep ensembles

Suppose we train M independently initialized models and average their predicted probabilities:

$$\bar{p}(x) = \frac{1}{M} \sum_{m=1}^{M} p^{(m)}(x).$$

This is the basic ensemble predictor.

Why ensembles often help under convex proper scores. Let $S(\cdot, y)$ be a convex score in the probability vector. Then Jensen's inequality gives

$$S(\bar{p}(x), y) = S\left(\frac{1}{M} \sum_{m=1}^{M} p^{(m)}(x), y\right) \leq \frac{1}{M} \sum_{m=1}^{M} S(p^{(m)}(x), y).$$

So for convex scores such as Brier score and log loss, the ensemble prediction can be no worse than the average individual prediction at the same example.

Operational interpretation. Different trained models tend to disagree most on ambiguous or fragile cases. Averaging smooths away idiosyncratic confidence spikes and often produces a stronger uncertainty signal.

7.3 MC dropout

At training time, dropout randomly zeros parts of the network. MC dropout [13] keeps dropout active at test time as well.

The basic procedure is:

1. for the same input x, run the network forward T times with different dropout masks;
2. obtain T probability vectors $p^{(1)}(x), \ldots, p^{(T)}(x)$;

3. average them:

$$\bar{p}(x) = \frac{1}{T}\sum_{t=1}^{T} p^{(t)}(x);$$

4. use the variability across the T samples as an uncertainty signal.

What a beginner should understand first. Operationally, MC dropout is a cheap way to simulate an ensemble without training many entirely separate models. It is not identical to a deep ensemble, but it serves a similar purpose: approximate prediction averaging.

What a more advanced reader should know. Gal and Ghahramani [13] interpret this procedure as approximate Bayesian inference with a Bernoulli variational family. That fuller ELBO derivation is given in Part II and Appendix B.

7.4 Predictive variance and mutual information

Given sampled probabilities $p^{(t)}(x)$, one often computes:

$$\bar{p}(x) = \frac{1}{T}\sum_{t=1}^{T} p^{(t)}(x),$$

predictive entropy

$$H(\bar{p}(x)) = -\sum_{k} \bar{p}_k(x) \log \bar{p}_k(x),$$

and the mutual-information style quantity

$$\mathrm{MI}(x) = H(\bar{p}(x)) - \frac{1}{T}\sum_{t=1}^{T} H(p^{(t)}(x)).$$

A high value of $\mathrm{MI}(x)$ indicates disagreement across sampled subnetworks, which is often used as a proxy for epistemic uncertainty.

7.5 A compact practical comparison

Method	Extra training cost	Test-time cost	Main strength / limitation
Temperature scaling	tiny	tiny	Excellent first baseline; repairs numeric confidence but not ranking.
Deep ensembles	high	moderate to high	Strong empirical uncertainty baseline; often best under shift; expensive.
MC dropout	low to moderate	moderate	Cheaper approximate model averaging; often weaker than ensembles but easy to retrofit.

Takeaway after the toolbox section

A strong beginner workflow is:

1. start with temperature scaling as the post-hoc baseline;
2. if reliability under shift or abstention quality matters a lot, compare against deep ensembles;
3. if training many models is too expensive, try MC dropout.

8 Selective prediction: when the model should abstain

Calibration tells us whether a confidence number is honest. Selective prediction uses confidence to decide *whether to answer at all*.

8.1 The simplest decision-theoretic derivation

Suppose the model may either:

- output a class label and incur loss 0 if correct and 1 if wrong, or
- abstain and incur a fixed reject cost $\lambda \in (0, 1)$.

Given input x, the best class to predict is the class with largest conditional probability:

$$k^\star(x) = \arg\max_k P(Y = k \mid X = x).$$

If we answer with that class, the conditional expected loss is

$$1 - \max_k P(Y = k \mid X = x).$$

If we abstain, the expected loss is simply λ.

Therefore we should answer exactly when

$$1 - \max_k P(Y = k \mid X = x) \leq \lambda.$$

Equivalently,

$$\max_k P(Y = k \mid X = x) \geq 1 - \lambda.$$

This is the most important beginner derivation in abstention: *answer only when the best-class probability is large enough to beat the abstention cost.*

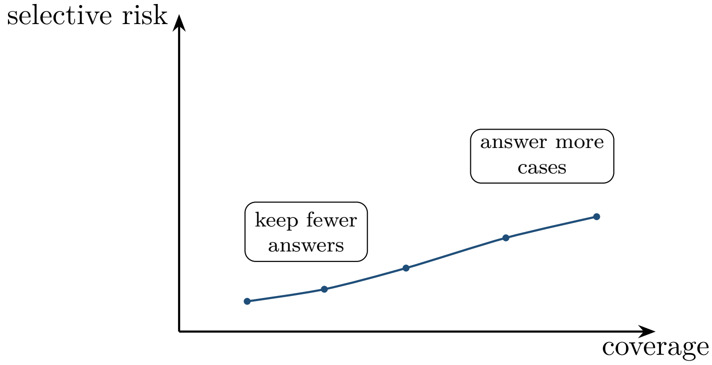

Figure 5: Risk–coverage behavior on the toy example. Moving left means answering on fewer cases but ideally making fewer mistakes among the accepted cases.

8.2 Coverage and selective risk

In practice we do not know the true conditional probabilities, so we use a confidence score $s(x)$ such as the top softmax probability. Given a threshold τ, the system answers on

$$\{x : s(x) \geq \tau\}$$

and abstains on the rest.

Two quantities are then fundamental:

Definition 8.1 (Coverage)**.** *Coverage is the fraction of examples on which the system answers:*

$$\mathrm{Cov}(\tau) = \mathbb{P}(s(X) \geq \tau).$$

Definition 8.2 (Selective risk)**.** *Selective risk is the error rate on the answered examples:*

$$R_{\mathrm{sel}}(\tau) = \mathbb{P}(\hat{Y} \neq Y \mid s(X) \geq \tau).$$

8.3 Risk–coverage curves

As we raise the threshold, coverage typically goes down and selective risk usually improves. Plotting one against the other gives the risk–coverage curve.

Figure 5 uses the confidence ranking of Example B.

How to read the curve. At coverage 1.0, the system answers every example, so selective risk is just the ordinary error rate. As we move left, the threshold increases, fewer examples are answered, and ideally the retained examples are more reliable.

8.4 Deployment-relevant reporting

A deployment-oriented report should usually include:

1. the score used for abstention;
2. the full risk–coverage curve;
3. selective risk at a few coverage targets such as 95%, 90%, and 80%;
4. the calibration quality of the *accepted* set if confidence is shown to users;
5. any distribution-shift stress test that changes the curve.

8.5 Why calibration and selective prediction are related but different

If confidence scores are miscalibrated, then a threshold such as "accept if confidence ≥ 0.9" can be numerically misleading. However, selective prediction also depends on *ranking quality*. A monotone recalibrator can fix numerical honesty without changing which examples are ranked above or below each other.

That is why the modern literature increasingly separates:

- **calibration quality:** are the numbers honest?
- **ranking quality:** do the hardest cases get the lowest scores?

8.6 A first mention of end-to-end reject-option methods

Some methods do not only threshold an existing confidence score. They train the network to learn answer-versus-abstain behavior directly. SelectiveNet [15] and Deep Gamblers [27] are the canonical examples. The formal optimization view appears in Part II. For a first read, it is enough to know that these methods make abstention part of the training objective rather than a post-processing step.

Takeaway after selective prediction

Calibration asks whether a confidence number is honest. Selective prediction asks what to do with that number. The basic reject rule is simple: answer if the expected mistake cost is smaller than the abstention cost.

Checkpoint questions

Can you now explain:

1. why the reject threshold is $1 - \lambda$ in the simplest cost model;
2. what coverage means;
3. what selective risk means;
4. why a monotone recalibrator may improve calibration but leave the risk–coverage curve unchanged?

9 Why calibration can break after deployment

A novice often asks: "If I calibrated on a validation set and the test set looked fine, why might the system still fail after deployment?"

The short answer is: because the data distribution can change.

9.1 Three common shift stories

Story 1: covariate shift. The label meanings stay the same, but the inputs look different. Examples: daytime images at training time and nighttime images at deployment; one hospital's scanner at training time and another hospital's scanner at deployment.

Story 2: label shift. The class-conditional input distributions stay similar, but class frequencies change. For example, flu prevalence may differ between seasons.

Story 3: out-of-distribution inputs. Deployment includes examples that were not represented in training. A classifier forced to choose among known classes can then be confidently wrong.

9.2 Why reliability is especially fragile under shift

Calibration is a conditional statement about the joint distribution of predictions and outcomes. If that joint distribution changes, calibration can change even when the model weights do not.

Historically, Ovadia *et al.* [37] and Minderer *et al.* [31] made this empirical lesson vivid: uncertainty estimates that look acceptable on clean IID test data can degrade sharply under shift.

9.3 The label-shift correction formula

Under *label shift*, one assumes

$$P_t(X \mid Y) = P_s(X \mid Y)$$

but class priors differ between source s and target t. Then Bayes' rule gives

$$P_t(Y = y \mid X = x) = \frac{w_y\, P_s(Y = y \mid X = x)}{\sum_j w_j\, P_s(Y = j \mid X = x)}, \qquad w_y = \frac{P_t(Y = y)}{P_s(Y = y)}.$$

This formula is worth remembering because it shows that a calibrator trained only on the source distribution may become wrong when class frequencies change.

9.4 A practical beginner lesson

A calibration report is much stronger if it includes:

- the validation/test reliability numbers on the nominal distribution,

- and at least one stress test under plausible shift.

Takeaway after the shift section

Good validation calibration does not guarantee good deployment calibration. Reliability must always be interpreted together with the distribution on which it was measured.

10 Core recap, self-test, and the bridge to deeper theory

At this point, a first-time reader should be able to say the following in plain language.

Eight things you should now be able to do

1. Explain the difference between logits, softmax probabilities, calibrated probabilities, and a decision threshold.
2. Explain the difference between accuracy, calibration, and sharpness.
3. Compute Brier score and log loss on a tiny binary table.
4. Compute ECE from binned confidence–correctness pairs.
5. Explain at least two reasons ECE can mislead.
6. Derive the temperature-scaling objective and explain why it preserves the top class.
7. Explain the operational difference between deep ensembles and MC dropout.
8. Derive the simplest abstention threshold and read a risk–coverage curve.

If one of these still feels shaky

Re-read only the corresponding sections of Part I. Do not move to Part II until the beginner core is comfortable. Part II is for deeper understanding, not for first contact.

Part II. Foundations and full derivations

How to use Part II

Part II is optional on a first read. Its purpose is to turn the intuitive story from Part I into a mathematically complete framework. The key design principle is: *state the problem, derive the target quantity, derive the final algorithm, and only then summarize the interpretation.*

11 Proper scoring rules in full generality

Part I already showed, in a binary setting, that Brier score and log loss are minimized by the true conditional probability. We now state the full multiclass framework.

11.1 Why proper scoring rules are the common theoretical root

Suppose the true conditional label distribution at input x is

$$q(x) = \big(q_1(x), \ldots, q_K(x)\big), \qquad q_k(x) = P(Y = k \mid X = x).$$

A scoring rule is a loss assigned to an announced probability vector $p \in \Delta^{K-1}$ and an observed label $y \in [K]$.

Definition 11.1 (Scoring rule)**.** *The conditional expected score of announcing p when the true distribution is q is*

$$\mathcal{R}_S(p; q) = \sum_{k=1}^{K} q_k \, S(p, k).$$

The rule is proper *if*

$$\mathcal{R}_S(q; q) \le \mathcal{R}_S(p; q) \quad \textit{for all } p, q,$$

and strictly proper *if equality holds only at $p = q$.*

This is the formal version of the beginner phrase "the loss rewards honest probabilities." If a learner minimizes a strictly proper score at the population level, the target it is trying to recover is the true conditional probability vector itself.

11.2 Full derivation: the multiclass log score is strictly proper

For multiclass classification, the log score is

$$S_{\log}(p, y) = -\log p_y.$$

Let q be the true label distribution. Then the conditional risk of announcing p is

$$\mathcal{R}_{\log}(p; q) = -\sum_{k=1}^{K} q_k \log p_k.$$

Now add and subtract $-\sum_k q_k \log q_k$:

$$\mathcal{R}_{\log}(p; q) = -\sum_{k=1}^{K} q_k \log q_k + \sum_{k=1}^{K} q_k \log \frac{q_k}{p_k}.$$

Recognize the two terms:

$$\mathcal{R}_{\log}(p; q) = H(q) + \mathrm{KL}(q \| p),$$

where

$$H(q) = -\sum_{k=1}^{K} q_k \log q_k$$

is the entropy and

$$\mathrm{KL}(q \| p) = \sum_{k=1}^{K} q_k \log \frac{q_k}{p_k} \geq 0$$

is the Kullback–Leibler divergence.

Because $\mathrm{KL}(q \| p) \geq 0$ with equality if and only if $p = q$, we obtain

$$\mathcal{R}_{\log}(p; q) \geq \mathcal{R}_{\log}(q; q),$$

with equality only when $p = q$. So the log score is strictly proper.

What the derivation teaches. The cross-entropy objective used throughout deep learning is not merely a convenient differentiable surrogate. At the population level it is entropy plus a KL penalty away from the true conditional distribution. That is why it is a canonical probability-estimation loss.

11.3 Full derivation: the multiclass Brier score is strictly proper

Let $e_Y \in \{0, 1\}^K$ denote the one-hot label vector. The multiclass Brier score is

$$S_{\mathrm{Br}}(p, Y) = \|p - e_Y\|_2^2.$$

Let $q = \mathbb{E}[e_Y \mid X = x]$. The conditional risk of announcing p is

$$\mathcal{R}_{\mathrm{Br}}(p; q) = \mathbb{E}[\|p - e_Y\|_2^2 \mid X = x].$$

Expand the square:

$$\|p - e_Y\|_2^2 = \|p\|_2^2 - 2p^\top e_Y + \|e_Y\|_2^2.$$

Take conditional expectation:

$$\mathcal{R}_{\mathrm{Br}}(p;q) = \|p\|_2^2 - 2p^\top q + \mathbb{E}[\|e_Y\|_2^2 \mid X = x].$$

Because e_Y is one-hot, $\|e_Y\|_2^2 = 1$, so

$$\mathcal{R}_{\mathrm{Br}}(p;q) = \|p\|_2^2 - 2p^\top q + 1.$$

Now add and subtract $\|q\|_2^2$:

$$\mathcal{R}_{\mathrm{Br}}(p;q) = \left(\|p\|_2^2 - 2p^\top q + \|q\|_2^2\right) + 1 - \|q\|_2^2.$$

Thus

$$\mathcal{R}_{\mathrm{Br}}(p;q) = \|p - q\|_2^2 + 1 - \|q\|_2^2.$$

The second term does not depend on p. Therefore the unique minimizer is

$$p = q.$$

So the Brier score is also strictly proper.

What the derivation teaches. For Brier score, the excess risk of announcing p instead of q is exactly the squared Euclidean distance $\|p - q\|_2^2$ plus a constant. That geometric form is why Brier score admits clean decompositions.

11.4 Murphy's decomposition in the binary case

Murphy's decomposition [32] separates uncertainty, resolution, and reliability. It remains one of the best mathematical explanations of why ECE alone is not enough.

Let $Y \in \{0, 1\}$ and let the model output a confidence $C = \hat{p}(X)$. Define

$$Q = \mathbb{E}[Y \mid C].$$

The binary Brier score is

$$\mathrm{BS} = \mathbb{E}[(Y - C)^2].$$

Insert and subtract Q:

$$Y - C = (Y - Q) + (Q - C).$$

Square:

$$(Y - C)^2 = (Y - Q)^2 + 2(Y - Q)(Q - C) + (Q - C)^2.$$

Take expectation. For the cross term,

$$\mathbb{E}[(Y - Q)(Q - C)] = \mathbb{E}[\mathbb{E}[(Y - Q)(Q - C) \mid C]]\,.$$

Since $(Q - C)$ is measurable with respect to C,

$$\mathbb{E}[(Y - Q)(Q - C) \mid C] = (Q - C)\mathbb{E}[Y - Q \mid C].$$

But

$$\mathbb{E}[Y - Q \mid C] = \mathbb{E}[Y - \mathbb{E}[Y \mid C] \mid C] = 0.$$

Hence the cross term vanishes, and

$$\mathrm{BS} = \mathbb{E}[(Y - Q)^2] + \mathbb{E}[(Q - C)^2].$$

The second term is the *reliability* term. It is zero exactly when $Q = C$, which is exactly calibration. Now simplify the first term. Conditionally on C, the label is Bernoulli with mean Q, so

$$\mathbb{E}[(Y - Q)^2 \mid C] = Q(1 - Q).$$

Therefore

$$\mathbb{E}[(Y - Q)^2] = \mathbb{E}[Q(1 - Q)].$$

Let $\bar{y} = \mathbb{E}[Y] = \mathbb{E}[Q]$. Then

$$\mathbb{E}[Q(1 - Q)] = \bar{y} - \mathbb{E}[Q^2].$$

Using

$$\mathrm{Var}(Q) = \mathbb{E}[Q^2] - \bar{y}^2 \quad \Longrightarrow \quad \mathbb{E}[Q^2] = \mathrm{Var}(Q) + \bar{y}^2,$$

we get

$$\mathbb{E}[Q(1 - Q)] = \bar{y} - \mathrm{Var}(Q) - \bar{y}^2 = \bar{y}(1 - \bar{y}) - \mathrm{Var}(Q).$$

Substituting back,

$$\mathrm{BS} = \underbrace{\bar{y}(1 - \bar{y})}_{\text{uncertainty}} - \underbrace{\mathrm{Var}(Q)}_{\text{resolution}} + \underbrace{\mathbb{E}[(Q - C)^2]}_{\text{reliability}}.$$

Interpretation. The decomposition says: a model can improve reliability by collapsing all forecasts toward the mean, but that may destroy resolution. This is exactly why modern papers insist that calibration must be reported together with proper scores.

11.5 What proper scoring rules do and do not guarantee

Strict propriety tells us the correct *population target*. It does *not* tell us that a finite network trained on finite data will be calibrated on a future test set. Miscalibration can still arise because of finite-sample error, model misspecification, optimization dynamics, regularization, and distribution shift. That is why post-hoc calibration, uncertainty estimation, and deployment auditing remain necessary.

12 Advanced post-hoc calibration families

12.1 The common template

Post-hoc calibration fits a map

$$g: \text{ old score object} \to \text{new probability object}$$

using a frozen base model and a separate calibration set. The main design choices are:

- the *input object*: a scalar score, a logit vector, or a full probability vector;
- the *function family*: parametric, monotone nonparametric, or density-inspired;
- the *objective*: usually NLL or a closely related proper score.

12.2 Platt scaling: the classical binary baseline

Platt scaling [38] takes a scalar score $s(x)$ and fits

$$g(s) = \text{sigmoid}(As + B) = \frac{1}{1 + e^{-(As+B)}}.$$

Given calibration pairs (s_i, y_i), the negative log-likelihood is

$$\mathcal{L}(A, B) = -\sum_{i=1}^{m} \left[y_i \log g(s_i) + (1 - y_i)\log(1 - g(s_i))\right].$$

There is no closed-form minimizer, so the final algorithm solves for (A, B) numerically with Newton's method, LBFGS, or another standard optimizer.

Gradient derivation. Write $g_i = g(s_i)$. Then

$$\frac{\partial \mathcal{L}}{\partial A} = \sum_{i=1}^{m}(g_i - y_i)s_i, \qquad \frac{\partial \mathcal{L}}{\partial B} = \sum_{i=1}^{m}(g_i - y_i).$$

This is the same gradient structure as ordinary logistic regression because Platt scaling is logistic regression on the scalar feature s.

Why it matters historically. Platt scaling is the ancestor of most later parametric post-hoc calibrators. Temperature scaling can be read as a very restricted multiclass analogue acting directly on logits.

12.3 Histogram binning and isotonic regression

Histogram binning partitions the score range into intervals and replaces each score by the empirical positive rate of its interval. It is simple but coarse.

Isotonic regression instead solves the monotone least-squares problem

$$\min_{a_1 \le a_2 \le \cdots \le a_m} \sum_{i=1}^{m} (a_i - y_i)^2,$$

after sorting the calibration examples by score. The fitted values (a_i) are constrained to be nondecreasing.

Why the pool-adjacent-violators (PAV) algorithm works. Suppose two adjacent blocks violate monotonicity: the left block has fitted value u and the right block has fitted value v with $u > v$. Then any feasible optimum must replace the two by a common value between them, because otherwise the monotonicity constraint is violated. The squared-loss minimizer for that merged block is its weighted average label. PAV repeatedly merges adjacent violating blocks and replaces them by block averages until no violations remain. That greedy merging procedure is exactly solving the constrained least-squares problem.

Strength and weakness. Isotonic regression is flexible and can fit complicated monotone distortions, but it can overfit when the calibration set is small.

12.4 Beta calibration

Beta calibration [21] extends logistic calibration by using the features

$$\log s, \qquad \log(1-s),$$

where $s \in (0,1)$ is the uncalibrated probability. Its functional form is

$$g(s) = \operatorname{sigmoid}\big(a \log s + b \log(1-s) + c\big).$$

Why these features arise. Suppose the class-conditional score densities are approximately Beta:

$$s \mid Y = 1 \sim \mathrm{Beta}(\alpha_1, \beta_1), \qquad s \mid Y = 0 \sim \mathrm{Beta}(\alpha_0, \beta_0).$$

Then the log posterior odds satisfy

$$\log \frac{P(Y=1 \mid s)}{P(Y=0 \mid s)} = \log \frac{\pi_1}{\pi_0} + (\alpha_1 - \alpha_0) \log s + (\beta_1 - \beta_0) \log(1-s) + \text{constant}.$$

So the natural discriminative model is affine in $\log s$ and $\log(1-s)$. This is the derivational reason for the beta-calibration family.

Why it improves on plain Platt scaling. Platt scaling uses only the raw score or logit. Beta calibration uses score features that can represent asymmetric distortions near 0 and 1. That makes it better suited to probabilities that are already in $(0,1)$ but not perfectly calibrated.

12.5 Dirichlet calibration

Dirichlet calibration [22] is a multiclass analogue operating on the full probability vector. Given

$$p = (p_1, \dots, p_K) \in \Delta^{K-1},$$

define

$$\phi(p) = \big(\log p_1, \dots, \log p_K\big).$$

Then fit a multinomial logistic regression model

$$g_k(p) = \frac{\exp(a_k^\top \phi(p) + b_k)}{\sum_{j=1}^K \exp(a_j^\top \phi(p) + b_j)}.$$

Why this is the natural multiclass extension. Beta calibration uses log features of a binary probability and its complement. The multiclass analogue is precisely the vector of log-probabilities. This lets the calibrator model interactions between classes that scalar temperature scaling cannot.

Temperature scaling as a special case. If the base probabilities come from logits z and we choose

$$a_k = \frac{1}{T} e_k, \qquad b_k = 0,$$

where e_k is the k-th standard basis vector, then

$$g_k(p) \propto \exp\left(\frac{1}{T} \log p_k\right) = p_k^{1/T}.$$

After renormalization, this is exactly temperature scaling. So temperature scaling is a restricted subfamily of Dirichlet-style vector calibration.

12.6 How the post-hoc family fits together

The family tree is:

$$\text{Platt} \to \text{Beta} \qquad \text{and} \qquad \text{Temperature scaling} \to \text{Dirichlet calibration}.$$

All of these methods differ mainly in the flexibility of the map g and in whether they recalibrate a scalar score or the full probability vector.

13 Deeper uncertainty-toolbox derivations

13.1 Deep ensembles under convex proper scores

Suppose we have probabilistic predictions $p^{(1)}, \dots, p^{(M)}$ and define

$$\bar{p} = \frac{1}{M} \sum_{m=1}^{M} p^{(m)}.$$

If the scoring rule $S(\cdot, y)$ is convex in its probability argument, then Jensen's inequality gives

$$S(\bar{p}, y) \leq \frac{1}{M} \sum_{m=1}^{M} S(p^{(m)}, y).$$

Taking expectation over (X, Y),

$$\mathbb{E}[S(\bar{p}(X), Y)] \leq \frac{1}{M} \sum_{m=1}^{M} \mathbb{E}[S(p^{(m)}(X), Y)].$$

This is the formal reason averaged predictors are attractive under convex proper scores such as log loss and Brier score.

The inequality does *not* prove that a practical deep ensemble must beat its best member on every metric. But it does explain why averaging is so effective as a baseline.

13.2 MC dropout: from Bayesian predictive averaging to a practical algorithm

Bayesian predictive ideal. A Bayesian neural network treats weights W as random. The predictive distribution is

$$p(y \mid x, \mathcal{D}) = \int p(y \mid x, W)\, p(W \mid \mathcal{D})\, dW.$$

This integral is intractable in large neural networks.

Variational approximation. Variational inference replaces the true posterior by a simpler family $q_\theta(W)$ and minimizes

$$\mathrm{KL}(q_\theta(W) \,\|\, p(W \mid \mathcal{D})).$$

Equivalently, it maximizes the evidence lower bound (ELBO)

$$\mathrm{ELBO}(\theta) = \mathbb{E}_{q_\theta(W)}[\log p(\mathcal{D} \mid W)] - \mathrm{KL}(q_\theta(W) \,\|\, p(W)).$$

Dropout variational family. Gal and Ghahramani [13] use Bernoulli random masks to define $q_\theta(W)$. Operationally, sampling a dropout mask corresponds to sampling one member of the variational family.

From ELBO to the practical objective. Approximating the expectation over dropout masks by Monte Carlo samples leads to the familiar training objective: cross-entropy on stochastic dropout forward passes plus weight decay. At test time, keeping dropout active and averaging repeated forward passes approximates

$$\int p(y \mid x, W)\, q_\theta(W)\, dW.$$

The final prediction rule. Draw T dropout masks, produce predictions $p^{(1)}(x), \ldots, p^{(T)}(x)$, and return

$$\bar{p}(x) = \frac{1}{T} \sum_{t=1}^{T} p^{(t)}(x).$$

The sample variance across the T draws estimates predictive uncertainty under the variational posterior.

Where the omitted algebra lives. Appendix B gives a more explicit ELBO derivation. The main point here is the chain:

$$\begin{aligned}\text{Bayesian predictive averaging} &\to \text{variational approximation with Bernoulli masks}\\ &\to \text{dropout training objective}\\ &\to \text{MC averaging at test time.}\end{aligned}$$

13.3 Predictive mean, variance, and mutual information

Given sampled class probabilities $p^{(t)}(x)$, define

$$\bar{p}(x) = \frac{1}{T} \sum_{t=1}^{T} p^{(t)}(x).$$

The predictive covariance of class probabilities is estimated by

$$\widehat{\mathrm{Cov}}(x) = \frac{1}{T} \sum_{t=1}^{T} \big(p^{(t)}(x) - \bar{p}(x)\big)\big(p^{(t)}(x) - \bar{p}(x)\big)^\top.$$

This matrix summarizes disagreement across sampled subnetworks. Predictive entropy and mutual information, already introduced in Part I, are scalar summaries of the same idea.

14 Advanced abstention methods and beyond-global calibration

14.1 SelectiveNet: end-to-end learning with a reject option

SelectiveNet [15] learns three coupled outputs: a prediction head $f_\theta(x)$, a selection head $g_\theta(x) \in [0, 1]$, and an auxiliary head used to stabilize training. The main optimization problem is a coverage-constrained selective risk objective:

$$\min_{\theta} \; R_{\text{sel}}(\theta) \quad \text{subject to} \quad \phi(\theta) \geq c,$$

where c is the desired coverage level and $\phi(\theta)$ is the empirical coverage induced by g_θ.

The selective risk term has the form

$$R_{\text{sel}}(\theta) = \frac{\frac{1}{n}\sum_{i=1}^{n} \ell(f_\theta(x_i), y_i)\, g_\theta(x_i)}{\phi(\theta)}, \qquad \phi(\theta) = \frac{1}{n}\sum_{i=1}^{n} g_\theta(x_i).$$

Because constrained optimization is inconvenient in raw form, the paper uses a penalty relaxation:

$$\mathcal{L}(\theta) = R_{\text{sel}}(\theta) + \lambda \max(0, c - \phi(\theta))^2 + \alpha\, \mathcal{L}_{\text{aux}}(\theta).$$

This is the final training algorithm: optimize the penalized objective jointly over the prediction and selection heads.

What the derivation teaches. SelectiveNet is not merely "threshold softmax later." It trains the network so that answer-versus-abstain behavior is part of the learned representation.

14.2 Deep Gamblers

Deep Gamblers [27] add an explicit reservation option to the label space. The model allocates probability mass across the ordinary classes plus a reservation output. Utility is paid for correct classification; the reserved mass acts like a bet held back from risky examples. The key conceptual move is to bake abstention into the probability simplex itself. This produces an end-to-end abstention mechanism without a separate post-hoc threshold.

14.3 Strong, local, and multicalibration

Global top-label calibration can miss serious failures. Three influential stronger notions are:

- **Local calibration** [28]: calibration should hold in local regions of score space or feature space, not only on global averages.
- **Multicalibration** [19]: calibration should hold simultaneously over many computationally identifiable subgroups.
- **Strong calibration testing** [11]: one should test whether there exists a subgroup with large residual miscalibration.

These notions are historically important because they changed the community's view from "a single global scalar summary" to "reliability may fail in small but important subpopulations."

14.4 Shift-aware calibration

Part I already gave the label-shift correction formula. The broader lesson is that post-hoc calibration is itself distribution dependent. Recent methods such as LaSCal [40] explicitly estimate target-domain shift and insert that estimate into the calibration step. The modern view is therefore:

$$\text{calibration problem} = \text{base model} + \text{calibration notion} + \text{deployment distribution}.$$

No one of these three pieces can be ignored.

Takeaway after Part II

Part II turns the intuitive story into a complete framework. Proper scoring rules define the target. Post-hoc methods choose a calibration family. Uncertainty toolkits approximate model averaging. Selective methods turn calibrated or uncertainty-aware scores into actions. And stronger notions of calibration remind us that global averages can hide operationally important failures.

Part III. Historical lineage and recent influential work

How to use Part III

Part III is a survey layer. Its purpose is not to introduce brand-new beginner concepts. Its purpose is to show how the same common framework grew historically and how recent papers extend it. Each recent-paper summary answers the same five questions:

1. What problem does the paper focus on?
2. Which part of the common framework does it modify?
3. What is the key mathematical move?
4. What does it improve over older work?
5. How does the paper support its claim: theorem, algorithm, empirical benchmark, or all three?

15 Historical lineage: from root to leaves

The field did not begin with deep learning. Its roots lie in weather forecasting, Bayesian statistics, and reject-option pattern recognition. The table below is a compact root-to-leaf map.

Year	Work	Why it matters	Core prerequisite
1950	Brier [2]	Introduced the Brier score, one of the canonical strictly proper scoring rules for probability forecasts.	Part I, Section 4
1957	Chow [4]	Early reject-option formulation for pattern recognition.	Part I, Section 8
1970	Chow [5]	Classic optimal error–reject trade-off; the root of abstention theory.	Part I, Section 8
1973	Murphy [32]	Decomposed Brier score into uncertainty, resolution, and reliability.	Part II, Section 11
1982	Dawid [6]	Formalized calibration from a Bayesian forecasting viewpoint.	Part I, Section 3
1983	DeGroot–Fienberg [7]	Clarified forecast evaluation and calibration in statistics.	Part I, Section 3
1999	Platt [38]	Logistic post-hoc calibration of SVM scores; the ancestor of many later parametric calibrators.	Part II, Section 12
2001	Zadrozny–Elkan [46]	Popularized histogram binning and isotonic-style calibration ideas in the binary setting.	Part II, Section 12
2002	Zadrozny–Elkan [47]	Extended score-to-probability ideas to multiclass settings.	Part II, Section 12
2005	Niculescu-Mizil–Caruana [35]	Landmark empirical comparison of probability quality across standard learning algorithms.	Part I, Sections 4–5
2007	Gneiting–Raftery [16]	Canonical modern treatment of strictly proper scoring rules.	Part II, Section 11

2010	El-Yaniv–Wiener [9]	Learning-theoretic foundations of selective classification.	Part I, Section 8
2015	Naeini *et al.* [33]	Bayesian binning as a practical calibration method.	Part II, Section 12
2016	Gal–Ghahramani [13]	Interpreted dropout as approximate Bayesian inference.	Part I, Section 7; Part II, Section 13
2017	Guo *et al.* [17]	Practical deep-learning landmark: neural networks are miscalibrated; temperature scaling is a strong baseline.	Part I, Section 6
2017	Lakshminarayanan *et al.* [24]	Deep ensembles as a simple, strong uncertainty baseline.	Part I, Section 7
2017	Kull *et al.* [21]	Beta calibration: a better-founded binary post-hoc family.	Part II, Section 12
2017	Geifman–El-Yaniv [14]	Practical selective classification for deep networks.	Part I, Section 8
2018	Kumar *et al.* [23]	Trainable kernel calibration metrics, connecting calibration to differentiable objectives.	Part III, recent-paper bridge
2018	Hébert-Johnson *et al.* [19]	Multicalibration: subgroup calibration as an algorithmic guarantee.	Part II, Section 14
2019	Geifman–El-Yaniv [15]	SelectiveNet: integrated reject option trained end to end.	Part II, Section 14
2019	Nixon *et al.* [36]	Careful critique of ECE variants and metric choices.	Part I, Section 5
2019	Ovadia *et al.* [37]	Landmark benchmark of uncertainty under shift.	Part I, Section 9
2019	Kull *et al.* [22]	Dirichlet calibration for multiclass post-hoc repair.	Part II, Section 12
2021	Minderer *et al.* [31]	Revisited modern neural-network calibration and its fragility.	Part I, Section 9
2022	Roelofs *et al.* [43]	Debiasing and more careful estimation of calibration error.	Part I, Section 5
2022	Luo *et al.* [28]	Local calibration: global averages can miss local failures.	Part II, Section 14
2022	Fisch *et al.* [12]	Calibrated selective classification: reliability of the accepted set.	Part II and Part III
2022	Popordanoska *et al.* [39]	Canonical calibration error estimator, a more principled continuous measure.	Part III, recent metrics section
2023–2025	Recent wave	Many-class calibration, truthfulness of calibration metrics, efficient testing, selective evaluation, label-shift calibration, and selective-performance limits.	Current section

A compact four-branch summary. The historical tree is easiest to remember as four intertwined branches.

1. **Forecast-evaluation branch:** Brier → Murphy → Dawid / DeGroot–Fienberg → Gneiting–Raftery.
2. **Post-hoc calibration branch:** Platt → Zadrozny–Elkan → beta / temperature / Dirichlet calibration.
3. **Uncertainty-approximation branch:** dropout as variational inference → deep ensembles →

reliability under shift.

4. **Reject-option branch:** Chow → selective classification → SelectiveNet and modern abstention losses.

Why this history matters. A great deal of apparently new deep-learning work is best understood as a modern expression of one of these older questions: How should a probability forecast be scored? How should it be repaired? How should we decide not to predict? What happens under shift or on subgroups?

16 Recent influential works (2023–2025)

This section is a curated recent-work survey, not an exhaustive bibliography. The papers below were chosen because they changed how researchers think about calibration metrics, many-class calibration, selective prediction, or deployment-aware reliability.

16.1 2023 bridge papers

16.1.1 Marx, Zalouk, and Ermon (NeurIPS 2023): calibration by distribution matching

Prerequisites. ECE intuition from Part I and the idea that binning is crude.

Problem. Can calibration be measured and optimized without relying on arbitrary bins?

Where it fits. This paper extends the *measurement and training-objective* branch rather than the classical one-parameter post-hoc branch [30].

Key mathematical move. The paper treats calibration as a *distribution matching* problem in a reproducing-kernel Hilbert space. Very roughly, it replaces a binned discrepancy by a smooth distance between two embedded distributions. The resulting objective is MMD-like:

$$\text{calibration mismatch} \rightsquigarrow \|\mu_{\text{forecast,label}} - \mu_{\text{ideal}}\|_{\mathcal{H}}^2.$$

This turns a hard-to-optimize binned statistic into a differentiable objective.

What it improves. It continues the MMCE line [23] but sharpens the distribution-matching viewpoint and reduces dependence on arbitrary bins.

How the paper supports the claim. The support is mainly algorithmic plus empirical: the paper defines a smooth trainable objective and shows competitive or improved calibration behavior on deep models [30].

16.1.2 Pugnana and Ruggieri (AISTATS 2023): AUC-based selective classification

Prerequisites. Coverage, selective risk, and the idea that abstention depends on ranking.

Problem. Many selective-classification summaries compress a whole curve into a scalar. What if the main question is *ranking quality* rather than average area?

Where it fits. This paper modifies the *evaluation objective* for selective prediction [41].

Key mathematical move. Selective prediction is fundamentally about sorting examples from safer to riskier. The paper develops AUC-style summaries that reward score functions that rank correct predictions above incorrect ones. The derivational pivot is:

$$\text{selective performance} \rightsquigarrow \text{pairwise ranking quality between correct and incorrect cases.}$$

What it improves. It makes explicit that calibration and ranking are different resources. A model can be well calibrated and still rank examples poorly for abstention.

How the paper supports the claim. The paper combines theoretical properties with experiments showing that AUC-style criteria can rank methods differently from traditional risk–coverage summaries [41].

16.1.3 Ding, Cao, and Luo (NeurIPS 2023): why deep ensembles help selective classification

Prerequisites. Deep ensembles from Part I and the idea that hard examples dominate abstention quality.

Problem. Why do deep ensembles often improve selective classification more than monotone post-hoc calibration does?

Where it fits. This paper connects the *uncertainty-estimation branch* to the *abstention branch* [8].

Key mathematical move. The paper identifies *top-ambiguity samples*—inputs near decision boundaries or with multiple plausible labels—as the cases on which ensemble disagreement is most useful. The practical derivation is conceptual:

$$\text{ensemble diversity} \to \text{better uncertainty on ambiguous points} \to \text{better ranking for abstention.}$$

What it improves. It explains why ensembles can improve risk–coverage curves even when a monotone calibrator would not.

How the paper supports the claim. The support is primarily empirical and diagnostic: controlled experiments isolate the benefit on the hard cases that dominate selective performance [8].

16.2 2024 papers on calibration metrics and auditing

16.2.1 Hu, Jambulapati, Tian, and Yang (NeurIPS 2024): testing calibration in nearly-linear time

Prerequisites. Population calibration, empirical calibration, and the idea of hypothesis testing.

Problem. Instead of merely estimating ECE, can we *test* calibration efficiently with algorithmic guarantees?

Where it fits. This paper targets the *measurement and testing* part of the framework [20].

Key mathematical move. The paper starts from a smooth-calibration testing linear program and shows that it can be reformulated as a minimum-cost flow problem on a structured graph:

$$\text{calibration testing LP} \Longrightarrow \text{minimum-cost flow} \Longrightarrow O(n \log^2 n)\text{-time algorithm.}$$

This is the central derivational step.

What it improves. It brings formal statistical testing language and near-linear-time algorithmics to calibration auditing.

How the paper supports the claim. The support is theorem-heavy: runtime guarantees, information-theoretic optimality, and experiments showing scale and power [20].

16.2.2 Feng, Gossmann, Pirracchio, Petrick, Pennello, and Sahiner (AISTATS 2024): testing for strong calibration

Prerequisites. Global calibration can hide subgroup failures.

Problem. Is the model reliable for *everyone*, or is there a subgroup with serious miscalibration?

Where it fits. This paper targets *subgroup reliability* rather than global calibration [11].

Key mathematical move. The paper introduces detector functions and rewrites subgroup detection as a structured residual search:

$$\sup_{h\in\mathcal{H}_+} \mathbb{E}[(Y - \hat{p}_\delta(X))h(X)].$$

A two-stage procedure then learns residual predictors, sorts examples by predicted residuals, and applies a cumulative-sum changepoint test. The conceptual derivation is:

$$\text{bad subgroup} \rightarrow \text{structured residual pattern} \rightarrow \text{detectable changepoint after sorting.}$$

What it improves. It is more powerful than naive subgroup binning when the problematic subgroup is small or subtle.

How the paper supports the claim. The support is both theoretical and empirical: finite-sample control of type-I error and stronger empirical power on simulated and medical-risk data [11].

16.2.3 Haghtalab, Qiao, Yang, and Zhao (NeurIPS 2024): truthfulness of calibration measures

Prerequisites. ECE is descriptive, not a proper score.

Problem. Should a calibration measure itself reward truthful prediction?

Where it fits. This paper questions the *optimization value* of popular calibration measures [18].

Key mathematical move. The paper defines a notion of *truthfulness*: truthful conditional forecasting should minimize the expected penalty, at least approximately. It then constructs distributions on which several existing calibration measures fail this property. To repair the issue, the paper proposes a smoothed subsampled measure (SSCE) for which truthful forecasting becomes approximately optimal.

What it improves. It adds a new criterion for judging calibration measures beyond convenience and consistency.

How the paper supports the claim. The support is mainly theoretical: separation examples, regret-style reasoning, and formal properties of the proposed measure [18].

16.2.4 Le Coz, Herbin, and Adjed (NeurIPS 2024): confidence calibration of classifiers with many classes

Prerequisites. Top-label versus full-vector calibration.

Problem. When the number of classes is very large, full-vector calibration can be data hungry. Can we target exactly the probability that the top class is correct?

Where it fits. This paper is a many-class refinement of the *top-label calibration* problem [25].

Key mathematical move. Let

$$U = c(X) = \max_k p_k(X), \qquad Z = \mathbf{1}\{Y = \hat{y}(X)\}.$$

Then confidence calibration becomes a binary calibration problem:

$$P(Z = 1 \mid U = u) = u.$$

This reduction is the main derivational idea: a many-class problem is recast as binary calibration of the event "the top prediction is correct."

What it improves. It uses calibration data more efficiently when the deployment quantity of interest is only the correctness probability of the displayed top prediction.

How the paper supports the claim. The support is algorithmic plus empirical: the paper builds many-class confidence calibrators and compares them on large-class image and text benchmarks [25].

16.2.5 Berta, Bach, and Jordan (AISTATS 2024): ROC-regularized isotonic regression

Prerequisites. Isotonic regression and the distinction between calibration and ranking.

Problem. Flexible calibrators such as isotonic regression may improve calibration but hurt discrimination or ranking.

Where it fits. This paper refines the *post-hoc calibration family* [1].

Key mathematical move. The paper augments isotonic calibration with a term that controls ROC behavior. In simplified form, the objective is of the type

$$\min_{g \text{ monotone}} \text{calibration loss}(g) + \lambda \cdot \text{ROC regularizer}(g).$$

The goal is to gain flexibility without sacrificing ranking as severely as unconstrained flexible calibration can.

What it improves. It directly acknowledges a practical tension that beginner readers should now recognize: calibration quality and ranking quality can pull in different directions.

How the paper supports the claim. The support is algorithmic plus empirical, with comparisons showing improved trade-offs between calibration and ROC-like discrimination criteria [1].

16.3 2024 papers on calibration and abstention methods

16.3.1 Esaki, Nakamura, Kawano, Tokuhisa, and Kutsuna (AISTATS 2024): accuracy-preserving calibration via statistical modeling on the probability simplex

Prerequisites. Temperature scaling preserves argmax; flexible vector calibrators may not.

Problem. Can we recalibrate the full multiclass vector while preserving top-label accuracy?

Where it fits. This paper belongs to the *multiclass post-hoc calibration* branch [10].

Key mathematical move. The paper models class probabilities on the simplex with a statistical family designed to preserve the top label. The core idea is:

$$\text{fit a richer distributional transform on the simplex} \quad \text{subject to preserving argmax decisions.}$$

This directly targets a weakness of several flexible vector calibrators.

What it improves. It seeks the flexibility of vector calibration without paying the usual price of accuracy degradation.

How the paper supports the claim. The support is mainly empirical plus modeling-based: comparisons on multiclass tasks show better calibration than scalar baselines while preserving accuracy [10].

16.3.2 Zollo, Deng, Snell, Pitassi, and Zemel (TMLR 2024): improving predictor reliability with selective recalibration

Prerequisites. Calibration and selective prediction are related but distinct.

Problem. Can recalibration be focused on the subset of predictions that a system will actually keep?

Where it fits. This paper sits exactly at the boundary between the *calibration* and *selective-prediction* branches [48].

Key mathematical move. The paper studies recalibration jointly with the accept/reject decision. Instead of calibrating all predictions equally, it chooses a selective region and calibrates with that downstream choice in mind. Conceptually:

$$\text{recalibration target} = \text{accepted region} + \text{confidence map}.$$

What it improves. It makes explicit that the relevant calibration problem may be the reliability of the *accepted* set, not the full set of examples.

How the paper supports the claim. The support is mainly algorithmic plus empirical, with gains on the reliability of retained predictions [48].

16.3.3 Mao, Mohri, and Zhong (AISTATS 2024): theoretically grounded losses for score-based multiclass abstention

Prerequisites. Reject-option learning and surrogate losses.

Problem. Which surrogate losses are statistically consistent for multiclass abstention?

Where it fits. This paper is a theory-heavy contribution to the *end-to-end abstention-loss* branch [29].

Key mathematical move. The paper derives losses whose minimizers recover Bayes-optimal abstaining decisions under score-based multiclass prediction. The main derivation is of the familiar consistency form:

$$\text{design surrogate loss} \to \text{show Bayes consistency} \to \text{derive implementable algorithm}.$$

What it improves. It gives a more principled loss-design story than ad hoc abstention penalties.

How the paper supports the claim. The support is both theoretical and empirical: consistency results plus experiments on benchmark datasets [29].

16.3.4 Narasimhan, Menon, Jitkrittum, and Kumar (ICLR 2024): plugin estimators for selective classification with OOD detection

Prerequisites. Selective prediction under deployment can include out-of-distribution points.

Problem. How should one abstain when some test examples may come from outside the training label space?

Where it fits. This paper broadens the *selective-prediction framework* to open-world deployment [34].

Key mathematical move. The Bayes-optimal open-world reject rule depends on multiple ingredients: in-distribution correctness probabilities, OOD probabilities, and score thresholds. The paper estimates each ingredient separately and plugs them into the optimal rule:

Bayes-optimal decision formula → estimate its components → plugin selective classifier.

What it improves. It moves the field beyond the closed-world assumption that every test point belongs to a training class.

How the paper supports the claim. The support is both theoretical and empirical: theory for the plugin estimators and experiments with OOD-contaminated evaluation [34].

16.3.5 Traub, Bungert, Lüth, Baumgartner, Maier-Hein, Maier-Hein, and Jäger (NeurIPS 2024): overcoming common flaws in selective-classification evaluation

Prerequisites. Risk–coverage curves and the fact that scalar summaries matter.

Problem. Do common scalar summaries such as AURC align with operational goals?

Where it fits. This is a major paper about the *evaluation methodology* of selective classification [44].

Key mathematical move. The paper defines generalized risk

$$R_{\text{gen}}(\tau) = \text{Cov}(\tau)\, R_{\text{sel}}(\tau),$$

which can be interpreted as the rate of undetected failures at threshold τ. Integrating generalized risk over thresholds yields AUGRC. The change is not cosmetic: it changes the operational meaning of the scalar summary.

What it improves. It provides a summary metric that the authors argue is more interpretable and task-aligned than older area summaries.

How the paper supports the claim. The support is empirical and methodological: large comparative benchmarks show that method rankings can change materially under AUGRC [44].

16.3.6 Popordanoska, Radevski, Tuytelaars, and Blaschko (NeurIPS 2024): LaSCal

Prerequisites. Label shift from Part I.

Problem. Can we recalibrate in a target domain under label shift even without target labels?

Where it fits. This paper extends the *post-hoc calibration branch under distribution shift* [40].

Key mathematical move. Under label shift,

$$P_t(Y = y \mid X = x) = \frac{w_y \, P_s(Y = y \mid X = x)}{\sum_j w_j \, P_s(Y = j \mid X = x)}, \qquad w_y = \frac{P_t(Y = y)}{P_s(Y = y)}.$$

LaSCal estimates the shift weights from unlabeled target predictions and inserts them into a calibration objective. That is the full conceptual derivation:

$$\text{shift identity} \rightarrow \text{estimate weights from target predictions} \rightarrow \text{shift-aware recalibration.}$$

What it improves. It is a clean example of calibration moving beyond the IID assumption.

How the paper supports the claim. The support is mostly empirical with a clear derivational basis from the label-shift identity [40].

16.4 2025 papers: reporting, theory, and selective-performance limits

16.4.1 Chidambaram and Ge (ICLR 2025): reassessing how to compare and improve the calibration of machine learning models

Prerequisites. Murphy-style thinking: calibration and sharpness must be reported together.

Problem. Why do some calibrators look strong under ECE but weak under proper scores?

Where it fits. This paper is about *how the field compares and reports calibrators* [3].

Key mathematical move. The paper develops a decomposition of Bregman-type losses into calibration and sharpness pieces, then uses that decomposition to motivate the calibration–sharpness diagram. The central message is:

$$\text{generalization quality} = \text{calibration component} + \text{sharpness / resolution component.}$$

What it improves. It gives the community a language for explaining why ECE-only comparisons can be misleading or trivially gamed.

How the paper supports the claim. The support is theorem plus empirical illustration: the decomposition is proved, estimators are analyzed, and examples show misleading method rankings under ECE alone [3].

16.4.2 Li and Sur (NeurIPS 2025): optimal and provable calibration in high-dimensional binary classification

Prerequisites. Platt scaling and the idea that high-dimensional asymptotics can change calibration behavior.

Problem. Can a post-hoc calibrator be proved calibrated and optimal in a realistic high-dimensional regime?

Where it fits. This paper is a theory advance in the *post-hoc calibration* branch [26].

Key mathematical move. The paper constructs an *angular calibration* predictor by interpolating an informative linear score with a chance component, where the interpolation weight depends on the angle between the estimated and true parameter vectors. The resulting predictor has the form

$$\hat{f}_{\mathrm{ang}}(\hat{w}^\top x; \hat{\theta}) = \mathbb{E}_Z \left[\sigma \left(\cos(\hat{\theta}) \cdot \frac{\hat{w}^\top x}{\|\hat{w}\|_\Sigma} + \sin(\hat{\theta}) \cdot Z \right) \right], \qquad Z \sim \mathcal{N}(0,1).$$

The paper proves asymptotic calibration and Bregman optimality in the studied regime. It also shows that Platt scaling converges to the same limit under suitable conditions.

What it improves. It is among the clearest rigorous links between a practical post-hoc calibrator and high-dimensional theory.

How the paper supports the claim. The support is theorem-heavy: asymptotic calibration, optimality guarantees, a consistent angle estimator, and empirical demonstrations [26].

16.4.3 Rabanser and Papernot (NeurIPS 2025): what does it take to build a performant selective classifier?

Prerequisites. Calibration quality and ranking quality are not the same.

Problem. Why is there often a large gap between a practical selective classifier and the oracle ranking that accepts correct predictions first?

Where it fits. This paper directly studies the *limit of post-hoc calibration for selective prediction* [42].

Key mathematical move. The paper defines a *selective-classification gap* and decomposes it into Bayes noise, approximation error, ranking error, statistical noise, and implementation or shift slack. The crucial observation for this note is: if g is monotone increasing, then

$$s_i \geq s_j \iff g(s_i) \geq g(s_j),$$

so post-hoc monotone calibration does not change the ranking. Therefore closing the gap requires methods that can *reorder* examples, not only rescale them.

What it improves. It gives a sharper answer to a question that practitioners had felt for years: why does temperature scaling sometimes improve confidence numbers but barely move selective performance?

How the paper supports the claim. The support is theorem plus empirical decomposition on synthetic and real benchmarks [42].

16.5 Synthesis of the recent literature

The recent wave does not replace the classical framework. It refines it in five directions.

1. **Metric realism.** ECE is not enough; truthfulness, bias, subgroup reliability, and efficient testing matter.
2. **Task specificity.** Top-label calibration, full-vector calibration, and abstention ranking are distinct problems.
3. **Shift awareness.** Calibration must be studied under label shift and OOD conditions, not only on nominal IID test sets.
4. **Selective integration.** Calibration and abstention are increasingly studied as one reliability problem rather than two separate ones.
5. **Ranking awareness.** For selective prediction, score ordering can matter more than global calibration.

Takeaway after Part III

The field's recent direction is not "abandon the classical view." It is "keep the classical view, but make it more realistic." That realism comes from better metrics, stronger subgroup notions, many-class formulations, shift-aware recalibration, and evaluation methods that care about ranking as well as numerical honesty.

17 Practical recipes for students and practitioners

This short section compresses the whole document into concrete default choices.

17.1 If you need one reliable workflow

1. Train the classifier with cross-entropy.
2. Report accuracy *and* a proper score (NLL or Brier).
3. Plot a reliability diagram and compute ECE with the binning rule stated explicitly.
4. Fit temperature scaling on a clean calibration split.
5. Compare against a deep-ensemble baseline if uncertainty quality matters.
6. If abstention matters, report the full risk–coverage curve and a few operating points.
7. Stress-test the system under at least one plausible deployment shift.

17.2 If the user will only see the top label and one confidence number

Favor top-label or confidence calibration diagnostics, but still report a proper score because a good-looking ECE alone can be misleading.

17.3 If the downstream system uses the full probability vector

Top-label ECE is not enough. Consider classwise or vector-aware calibration diagnostics and, if necessary, vector calibrators such as Dirichlet calibration.

17.4 If selective prediction is part of deployment

Remember the distinction between numerical honesty and ranking quality. Temperature scaling may improve the first without improving the second. Test the score function itself on the risk–coverage curve.

17.5 If deployment shift matters

Treat calibration as distribution specific. Source-domain temperature scaling is only a first step. Label-shift or OOD-aware methods become relevant when the deployment distribution genuinely changes.

18 Conclusion

The field of reliable classification is easiest to understand as a chain.

probability estimation $\to$ *probability scoring* $\to$ *post-hoc repair* $\to$ *uncertainty-aware prediction* $\to$ *answer / abstain decision.*

Proper scoring rules form the theoretical root because they tell us what honest probability estimation means. ECE is useful but limited because it is a descriptive summary rather than a proper objective. Temperature scaling is the practical workhorse because it is easy to fit and easy to explain. Deep ensembles and MC dropout strengthen the uncertainty signal beyond single-model softmax confidence. Selective prediction then asks what to do with that signal, and deployment shift reminds us that reliability is always distribution dependent.

For a novice, the most important lesson is not a formula. It is a workflow: start with a clean split, score probabilities with principled losses, audit calibration with more than one diagnostic, and evaluate the final decision rule that will actually be used. For an advanced reader, the broader lesson is that the field is converging on a richer notion of reliability that combines calibration, sharpness, ranking quality, subgroup behavior, and shift robustness.

A Appendix A: full calculations for the two toy examples

A.1 Softmax and summary statistics for the multiclass toy example

For the eight examples in Table 1:

$$\text{accuracy} = \frac{4}{8} = 0.5, \qquad \text{mean top confidence} \approx 0.681.$$

The multiclass Brier score is

$$\text{BS} = \frac{1}{8}\sum_{i=1}^{8}\sum_{k=1}^{3}(p_{i,k} - e_{y_i,k})^2 \approx 0.461,$$

and the average negative log-likelihood is

$$\text{NLL} = -\frac{1}{8}\sum_{i=1}^{8}\log p_{i,y_i} \approx 0.724.$$

A.2 ECE calculation for the multiclass toy example

With three equal-width bins

$$[0, \tfrac{1}{3}),\ [\tfrac{1}{3}, \tfrac{2}{3}),\ [\tfrac{2}{3}, 1],$$

the nonempty bins are:

Bin	Confidences	Correctness	Bin size	Mean conf.	Mean acc.
$[\frac{1}{3}, \frac{2}{3})$	$0.595, 0.426, 0.500$	$0, 0, 0$	3	0.507	0.000
$[\frac{2}{3}, 1]$	$0.894, 0.696, 0.777, 0.812, 0.746$	$1, 1, 1, 1, 0$	5	0.785	0.800

Hence

$$\text{ECE} = \frac{3}{8}|0.000 - 0.507| + \frac{5}{8}|0.800 - 0.785| \approx 0.199.$$

A.3 Why Example C is useful

Model A and Model B have the same accuracy, namely 5/8. But they differ on probability-quality metrics:

$$\text{BS}_A = 0.2075 < 0.2213 = \text{BS}_B, \qquad \text{NLL}_A \approx 0.589 < 0.633 \approx \text{NLL}_B.$$

Under one coarse two-bin ECE scheme, Model B can nevertheless obtain ECE $= 0$. This is a compact numerical demonstration that ECE alone can be misleading.

B Appendix B: a more explicit ELBO derivation for MC dropout

B.1 Bayesian predictive distribution

A Bayesian neural network defines

$$p(y \mid x, \mathcal{D}) = \int p(y \mid x, W)\, p(W \mid \mathcal{D})\, dW.$$

Since the posterior $p(W \mid \mathcal{D})$ is intractable, approximate it by a variational family $q_\theta(W)$.

B.2 Deriving the ELBO

Start from the identity

$$\log p(\mathcal{D}) = \text{ELBO}(\theta) + \text{KL}(q_\theta(W) \,\|\, p(W \mid \mathcal{D})),$$

where

$$\text{ELBO}(\theta) = \mathbb{E}_{q_\theta(W)}[\log p(\mathcal{D} \mid W)] - \text{KL}(q_\theta(W) \,\|\, p(W)).$$

Because KL divergence is nonnegative, maximizing ELBO minimizes the distance to the true posterior.

B.3 Bernoulli masks as the variational family

MC dropout chooses a family in which each weight block is multiplied by a Bernoulli random variable. Sampling a dropout mask therefore samples one effective weight realization from $q_\theta(W)$.

B.4 Approximating the ELBO in practice

The expectation

$$\mathbb{E}_{q_\theta(W)}[\log p(\mathcal{D} \mid W)]$$

is approximated by stochastic forward passes with dropout. The KL term becomes a weight-decay-like regularizer under the approximations used in [13]. This yields the familiar practical objective: cross-entropy under dropout noise plus weight decay.

B.5 Test-time prediction

At test time, keep dropout active, draw T masks, compute

$$p^{(t)}(y \mid x), \qquad t = 1, \ldots, T,$$

and average:

$$\bar{p}(y \mid x) = \frac{1}{T} \sum_{t=1}^{T} p^{(t)}(y \mid x).$$

This is the approximate posterior predictive distribution induced by the variational family.

C Appendix C: papers that are important even when they are mostly conceptual or empirical

Not every important paper in this area introduces a new algorithm with a long derivation. Some papers matter because they changed the community's mental model.

- **Guo *et al.* (2017)** [17] is historically crucial because it made neural-network miscalibration vivid and popularized temperature scaling.
- **Niculescu-Mizil and Caruana (2005)** [35] is historically important because it compared probability quality across methods in a systematic way.
- **Ovadia *et al.* (2019)** [37] is crucial because it benchmarked uncertainty under shift and changed practical expectations.
- **Minderer *et al.* (2021)** [31] strengthened the empirical message that calibration and uncertainty degrade in modern settings.

In such cases the right pedagogical move is not to force a nonexistent derivation. It is to explain the experimental design, the conceptual move, and the field-level consequence.

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